

Practical 8

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In [12]: # — Hill Cipher —  
import numpy as np  
import string
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In [13]: # Step 1: Key Matrix & Modular Inverse (for decryption)  
def get_key_matrix(keyword, n):  
    keyword = keyword.lower().replace(' ', '')  
    key = [ord(c) - 97 for c in keyword[:n*n]]  
    return np.array(key).reshape(n, n)  
  
def mod_inverse(a, m=26):  
    for x in range(1, m):  
        if (a * x) % m == 1:  
            return x  
  
def matrix_mod_inverse(matrix, mod=26):  
    det = int(round(np.linalg.det(matrix))) % mod  
    det_inv = mod_inverse(det % mod)  
    adjugate = np.round(det * np.linalg.inv(matrix)).astype(int) % mod  
    return (det_inv * adjugate) % mod
```

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In [14]: # Step 2: Prepare Text (pad to multiple of n)  
def prepare_text(text, n):  
    text = text.lower().replace(' ', '')  
    text = ''.join(c for c in text if c.isalpha())  
    while len(text) % n != 0:  
        text += 'x' # pad with 'x' if needed  
    return text
```

```
In [15]: # Step 3: Encrypt / Decrypt  
def hill(text, keyword, mode='encrypt'):  
    n = int(len(keyword.replace(' ', '')) ** 0.5)  
    key_matrix = get_key_matrix(keyword, n)  
    text = prepare_text(text, n)  
  
    if mode == 'decrypt':  
        key_matrix = matrix_mod_inverse(key_matrix).astype(int)  
  
    result = ""  
    for i in range(0, len(text), n):  
        block = np.array([ord(c) - 97 for c in text[i:i+n]])  
        encoded = np.dot(key_matrix, block) % 26  
        result += ''.join(chr(int(c) + 97) for c in encoded)  
  
    return result  
  
# — Demo —  
keyword = "gybnqkurp" # 3x3 key matrix (9 letters)
```

```
msg      = "Hello"

enc = hill(msg, keyword, mode='encrypt')
dec = hill(enc, keyword, mode='decrypt')

print("=== Hill Cipher ===")
print(f"Keyword   : {keyword}")
print(f"Key Matrix:\n{get_key_matrix(keyword, 3)}")
print(f"Original   : {msg}")
print(f"Encrypted   : {enc}")
print(f"Decrypted   : {dec}")
```

```
=== Hill Cipher ===
Keyword   : gybnqkurp
Key Matrix:
[[ 6 24  1]
 [13 16 10]
 [20 17 15]]
Original   : Hello
Encrypted   : tfjjzx
Decrypted   : ghnkvn
```